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SECONDARY BATTERY

Abstract

A secondary battery includes an electrode assembly including a cathode and an anode, a case in which the electrode assembly is housed, and a cap assembly which covers the case. The cap assembly includes a cap plate which covers the case and includes a terminal hole formed therein, a gasket disposed on the cap plate around the terminal hole, and a rivet which is inserted into the terminal hole to be coupled to the cap plate through the gasket, and includes a first protrusion part inserted into an upper portion of the gasket.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Korean Patent Application No. 10-2024-0030973 filed on Mar. 4, 2024 and Korean Patent Application No. 10-2024-0078331 filed on Jun. 17, 2024 in the Korean Intellectual Property Office (KIPO), the entire disclosure of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The embodiments of the present application relate to a secondary battery, and more specifically, to a secondary battery which includes a case and an electrode assembly.

2. Description of the Related Art

[0003] A secondary battery is a battery which can be repeatedly charged and discharged. With rapid progress of information and communication, and display industries, the secondary battery has been widely applied to various portable electronic telecommunication devices such as a camcorder, a mobile phone, a laptop computer as a power source thereof. Recently, a battery pack including the secondary battery has also been developed and applied to an eco-friendly automobile such as an electric vehicle, etc., as a power source thereof.

[0004] Examples of the secondary battery may include a lithium secondary battery, a nickel-cadmium battery, a nickel-hydrogen battery and the like. Among them, the lithium secondary battery has a high operating voltage and a high energy density per unit weight, and is advantageous in terms of a charging speed and light weight, such that development thereof has been proceeded in this regard.

[0005] The secondary battery may include an electrode assembly including a cathode, an anode and a separation membrane disposed between the cathode and the anode. The electrode assembly may be housed in an external case, such as a pouch, a circular case, a square case, etc., together with an electrolyte, to define a battery cell.

[0006] The case may include terminals for supplying current to the electrode assembly. In a terminal coupling structure for connecting the terminals and the electrode assembly, chemical and mechanical defects may be caused through an interface of the components.

SUMMARY OF THE INVENTION

[0007] An object of the present disclosure is to provide a secondary battery having improved mechanical stability and operational reliability.

[0008] A secondary battery according to an aspect of the present disclosure includes: an electrode assembly which includes a cathode and an anode; a case in which the electrode assembly is housed; and a cap assembly which covers the case. The cap assembly includes: a cap plate which covers the case and includes a terminal hole formed therein; a gasket disposed on the cap plate around the terminal hole; and a rivet which is inserted into the terminal hole to be coupled to the cap plate through the gasket. The rivet includes a first protrusion part inserted into an upper portion of the gasket.

[0009] In some embodiments, the rivet may include a rivet body inserted into the terminal hole and a rivet wing extending from the rivet body in a horizontal direction, wherein the first protrusion part may extend downward from a lower surface of the rivet wing.

[0010] In some embodiments, a separation distance between the first protrusion part and the rivet body may be 30% or more of a length of the rivet wing.

[0011] In some embodiments, the gasket may include a vertical extension part which is in contact with an outer circumferential surface of the rivet body, and a gasket wing extending from the vertical extension part in the horizontal direction, wherein the first protrusion part is inserted into an upper portion of the gasket wing.

[0012] In some embodiments, the cap plate may include a second protrusion part inserted into a lower portion of the gasket wing.

[0013] In some embodiments, the second protrusion part extends upward from an upper surface of the cap plate.

[0014] In some embodiments, a plurality of the first protrusion parts and a plurality of the second protrusion parts are inserted into the gasket wing.

[0015] In some embodiments, the first protrusion parts and the second protrusion parts are arranged alternately in the horizontal direction.

[0016] In some embodiments, any one second protrusion part of the second protrusion parts may be overlapped with two or more first protrusion parts in a vertical direction.

[0017] In some embodiments, a distance of the second protrusion part from an outer circumferential surface of the gasket wing may be 15% or more of a length of the gasket wing.

[0018] In some embodiments, the cap assembly may further include an upper insulation part disposed on the cap plate to partially cover the gasket wing.

[0019] In some embodiments, the cap assembly may further include a lower insulation part disposed on a lower surface of the cap plate to be in contact with the vertical extension part of the gasket.

[0020] A secondary battery according to another aspect of the present disclosure includes: an electrode assembly which includes a cathode and an anode; a case in which the electrode assembly is housed; and a cap assembly which covers the case. The cap assembly includes: a cap plate which covers the case and includes a terminal hole formed therein; a gasket which is disposed on the cap plate around the terminal hole, and includes a vertical extension part and a gasket wing bent from the vertical extension part in a horizontal direction; and a rivet which is coupled to the cap plate through the gasket, and includes a rivet body inserted into the terminal hole and a rivet wing extending from the rivet body in the horizontal direction. The rivet wing or the cap plate includes a protrusion part which is engaged to the gasket wing.

[0021] In some embodiments, the gasket wing may be sandwiched between the rivet wing and the cap plate.

[0022] In some embodiments, the vertical extension part and the gasket wing may be a member integrally formed with each other.

[0023] In some embodiments, the protrusion part may include a first protrusion part protruding downward from a lower surface of the rivet wing.

[0024] In some embodiments, the protrusion part may include a second protrusion part protruding upward from an upper surface of the cap plate.

[0025] In some embodiments, the case may be a square case.

[0026] In some embodiments, the cap assembly may further include a terminal plate disposed on the rivet.

[0027] According to the above-described embodiments, the gasket and the rivet of the cap assembly of the secondary battery may include wing parts which extend in the horizontal direction, respectively. Accordingly, inflow of external foreign substances and electrolyte leakage may be effectively blocked, while increasing coupling stability between the gasket and the rivet.

[0028] According to the embodiments of the present disclosure, the rivet wing may include a protrusion part inserted into an upper portion of the gasket wing. In some embodiments, the cap plate of the cap assembly may include a protrusion part inserted into a lower portion of the gasket wing. Due to the protrusion parts, it is possible to more effectively block electrolyte leakage and enhance the stability of the terminal coupling structure.

[0029] The secondary battery manufactured using an electrolyte injection device of the present disclosure may be widely applied to green technology fields such as an electric vehicle, and a battery charging station, as well as other solar power generation and wind power generation using the batteries. The secondary battery may be used in an eco-friendly electric vehicle, and a hybrid vehicle, etc., which are intended to prevent climate change by suppressing air pollution and greenhouse gas emissions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] The above and other objects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0031] FIG. 1 is a schematic perspective view illustrating a secondary battery according to exemplary embodiments;

[0032] FIG. 2 is a schematic partial cross-sectional view illustrating the secondary battery according to exemplary embodiments;

[0033] FIGS. 3 and 4 are partial enlarged cross-sectional views illustrating a terminal coupling structure of a cap assembly according to exemplary embodiments;

[0034] FIGS. 5 and 6 are partial enlarged cross-sectional views illustrating a terminal coupling structure of the cap assembly according to exemplary embodiments;

[0035] FIGS. 7 and 8 are partial enlarged cross-sectional views illustrating terminal coupling structures of the cap assembly according to exemplary embodiments; and

[0036] FIGS. 9 to 11 are partial enlarged cross-sectional views illustrating terminal coupling structures of the cap assembly according to exemplary embodiments.

DETAILED DESCRIPTION OF THE INVENTION

[0037] Exemplary embodiments provided through the present disclosure will be described in more detail with reference to the accompanying drawings. However, the drawings and embodiments attached to the present specification serve to further understand the technical spirit of the present disclosure, such that the present disclosure should not be construed as being limited only to the matters illustrated and described in these drawings and embodiments.

[0038] As used herein, the terms indicating space/locations such as a “lower portion,” an “upper portion,” “below” and “upper” are used in a relative sense to distinguish between components, and do not limit the absolute order or locations.

[0039] The term “connected” as used herein encompasses the case where two different components are directly connected, and the case where these components are indirectly connected through an intermediate component.

[0040] FIG. 1 is a schematic perspective view illustrating a secondary battery according to exemplary embodiments. FIG. 2 is a schematic partial cross-sectional view illustrating the secondary battery according to exemplary embodiments. For example, FIG. 2 is a partial cross-sectional view taken on line I-I' of FIG. 1 in a third direction.

[0041] In FIGS. 1 and 2, the first direction, the second direction and the third direction may be a longitudinal direction, a width direction and a height direction of the secondary battery or a case 110, respectively. A direction opposite to the direction indicated by the arrow may also be described as the same direction. The definition of the directions may be substantially equally applied to the remaining drawings.

[0042] Referring to FIGS. 1 and 2, a secondary battery 100 may include the case 110 and an electrode assembly 50 housed in the case 110.

[0043] As shown in FIG. 1, the case 110 may have a square cylinder shape, for example, in the

form of a hexagonal column. However, the shape of the case **110** may be appropriately changed in consideration of subjects to which the secondary batteries are applied, the desired capacity/output and the like. For example, the case **110** may have a circular cylinder shape, a coin shape, etc. [0044] The case **110** may include a first side surface **110a** and a second side surface **110b** which face each other in the first direction. The case **110** may include a front surface **110c** and a rear surface **110d** which face each other in the second direction. The case **110** may include a lower surface **110e** included at one end in the third direction, and a cap assembly **190** may be coupled or assembled to an upper portion of the case **110** in the third direction. The upper portion of the case **110** may be closed or sealed by the cap assembly **190**.

[0045] The electrode assembly **50** may include repeatedly stacked cathodes **70** and anodes **60**. In some embodiments, the cathodes **70** and the anodes **60** may be alternately and repeatedly stacked in the third direction with a separation membrane **80** interposed therebetween.

[0046] The cathode **70** may include a cathode current collector **75** and cathode active material layers **72** coated on surfaces (e.g., an upper surface and a lower surface in the third direction) of the cathode current collector **75**. The cathode current collector **75** may include stainless steel, nickel, aluminum, titanium, or an alloy thereof.

[0047] The cathode active material layer **72** includes a lithium-containing cathode active material, and the secondary battery **100** may be provided as a lithium secondary battery.

[0048] The lithium-containing cathode active material may include lithium-nickel oxide, lithium-cobalt oxide, lithium-manganese oxide, lithium-nickel-cobalt-manganese oxide, lithium iron phosphate compound, lithium-nickel-aluminum oxide, lithium-cobalt-aluminum oxide, etc.

[0049] The anode **60** may include an anode current collector **65** and anode active material layers **62** coated on surfaces (e.g., an upper surface and a lower surface in the third direction) of the anode current collector **65**. The anode current collector **65** may include copper, stainless steel, nickel, titanium, or an alloy thereof.

[0050] The anode active material layer **62** may include a carbon-based active material such as natural graphite or artificial graphite as an anode active material and/or a silicon-containing active material (e.g., SiO_x ($0 < x < 2$)). The anode active material may also include a silicon-carbon composite material.

[0051] The separation membrane **80** may include a porous polymer film made of a polyolefin polymer such as ethylene homopolymer, propylene homopolymer, ethylene/butene copolymer, ethylene/hexene copolymer, ethylene/methacrylate copolymer, etc. The separation membrane **80** may also include a nonwoven fabric made of glass fiber having a high melting point, polyethylene terephthalate fiber, etc.

[0052] The separation membrane **80** may also include a ceramic-based material. For example, inorganic particles may be coated on the polymer film or dispersed in the polymer film to improve heat resistance.

[0053] The cathodes **70** and the anodes **60** may be repeatedly and alternately laminated with the separation membrane **80** interposed therebetween by winding, stacking, zigzag-folding, etc. of the separation membrane **80**, thus to form the electrode assembly **50**. In one embodiment, the electrode assembly **50** may be prepared in a jelly-roll shape.

[0054] One end of the cathode current collector **75** included in the cathode **70** of each layer may be provided as a cathode tab **77**. One end of the anode current collector **65** included in the anode **60** of each layer may be provided as an anode tab **67**.

[0055] The cathode tabs **77** may protrude to one side (e.g., adjacent to the second side surface **110b**) in the first direction and be collected, thus to be connected to a cathode lead **92**. The anode tabs **67** may protrude to the other side (e.g., adjacent to the first side surface **110a**) in the first direction and be collected, thus to be connected to an anode lead **95**.

[0056] In some embodiments, the cathode tabs **77** may be fused to each other and connected to the cathode lead **92**. In one embodiment, the cathode tabs **77** may be fused together with the cathode

lead **92**. In some embodiments, the anode tabs **67** may be fused together and connected to the anode lead **95**. In one embodiment, the anode tabs **67** may be fused together with the anode lead **95**.

[0057] In FIG. **2**, it is illustrated that the cathodes **70** and the anodes **60** are stacked in the third direction and connected to the electrode leads on both sides in the first direction. However, the stacking direction and the electrode lead connection positions of the cathodes **70** and the anodes **60** may be appropriately changed in consideration of the size of the case **110** and the desired capacity/output of the battery.

[0058] In some embodiments, the electrode assembly **50** may be housed within the case **110** together with an electrolyte. A non-aqueous electrolyte may be used as the electrolyte.

[0059] The non-aqueous electrolyte includes a lithium salt of an electrolyte and an organic solvent, the lithium salt is represented by, for example, Li^+X^- , and as an anion (X^-) of the lithium salt, F^- , Cl^- , Br^- , I^- , NO_3^- , $\text{N}(\text{CN})_2^-$, BF_4^- , ClO_4^- , PF_6^- , $(\text{CF}_3)_2\text{PF}_4^-$, $(\text{CF}_3)_3\text{PF}_3^-$, $(\text{CF}_3)_3\text{PF}_2^-$, $(\text{CF}_3)_3\text{PF}^-$, $(\text{CF}_3)_3\text{SO}_3^-$, $(\text{CF}_3)_3\text{CF}_2\text{SO}_3^-$, $(\text{CF}_3)_3\text{SO}_2\text{N}^-$, $(\text{FSO}_2)_2\text{N}^-$, $\text{CF}_3\text{CF}_2(\text{CF}_3)_2\text{CO}^-$, $(\text{CF}_3)_3\text{SO}_2\text{CH}_3^-$, $(\text{SF}_5)_3\text{C}^-$, $(\text{CF}_3)_3\text{SO}_2\text{C}^-$, $\text{CF}_3(\text{CF}_2)_7\text{SO}_3^-$, CF_3CO_2^- , CH_3CO_2^- , SCN^- and $(\text{CF}_3)_3\text{CF}_2\text{SO}_2\text{N}^-$, etc. may be exemplified.

[0060] As the organic solvent, for example, propylene carbonate (PC), ethylene carbonate (EC), diethyl carbonate (DEC), dimethyl carbonate (DMC), ethylmethyl carbonate (EMC), methylpropyl carbonate, dipropyl carbonate, dimethylsulfoxide, acetonitrile, dimethoxyethane, diethoxyethane, vinylene carbonate, sulfolane, gamma-butyrolactone, propylene sulfite and tetrahydrofuran, etc. may be used. These may be used alone or in combination of two or more thereof.

[0061] The electrode leads (the cathode lead **92** and the anode lead **95**) may be connected to electrode terminal parts **112** and **115** included in the cap assembly **190**, respectively. The cathode lead **92** may be connected to a cathode terminal part **112**, and the anode lead **95** may be connected to an anode terminal part **115**.

[0062] The cap assembly **190** includes a cap plate **180** exposed to an outside of the secondary battery **100**, and the case **110** in which the electrode assembly **50** is housed may be sealed by the cap plate **180**.

[0063] The cap plate **180** may further include a vent part **183** and an electrolyte injection part **181**. The vent part **183** and the electrolyte injection part **181** include a vent hole and an electrolyte injection hole, respectively, and may include a plug or plate configured to seal the vent hole and the electrolyte injection hole.

[0064] The vent part **183** and the electrolyte injection part **181** may be formed between the cathode terminal part **112** and the anode terminal part **115**.

[0065] The electrode terminal parts **112** and **115A** may have a terminal coupling structure formed therein to be electrically connected or coupled with the electrode leads **92** and **95**, respectively. Configurations and structure of the cap assembly **190** and the terminal coupling structure will be described in more detail below.

[0066] FIGS. **3** and **4** are schematic partial enlarged cross-sectional views illustrating a terminal coupling structure of a cap assembly according to exemplary embodiments. For example, FIG. **3** is a cross-sectional view of the terminal coupling structure taken on line I-I' of FIG. **1** in the third direction. FIG. **4** is a partial enlarged cross-sectional view of an area A of FIG. **3**.

[0067] Referring to FIG. **3**, the terminal coupling structure included in the cap assembly **190** may include a terminal plate **130**, an upper insulation part **140**, a rivet **150** and a gasket **160**. The terminal plate **130** may be exposed to the outside of the secondary battery **100** to be provided as a substantial external terminal. A terminal post **120** may be inserted into a central portion of the

terminal plate **130**.

[0068] The upper insulation part **140** may include an insulation material such as a resin material. The upper insulation part **140** may be disposed on an upper surface of the cap plate **180** to support and fix a lower portion of the terminal plate **130**. The terminal plate **130** and the cap plate **180** may be insulated from each other by the upper insulation part **140**.

[0069] The rivet **150** may be inserted by penetrating the terminal plate **130** and the upper insulation part **140**. An upper portion of the rivet **150** may be in contact or coupled with the terminal post **120**. The terminal plate **130** and the upper insulation part **140** may be welded or fastened to the cap assembly **190** through the upper portion of the rivet **150**.

[0070] The terminal plate **130**, the terminal post **120** and the rivet **150** may include a conductive material such as metals or an alloy thereof.

[0071] The gasket **160** may fix the rivet **150** to the cap plate **180**. The gasket **160** may seal a space between the cap plate **180** and the rivet **150**. The gasket **160** may include an insulation resin material such as a perfluoroalkoxy (PFA) resin.

[0072] The cap plate **180** may include a terminal hole **182**. A gasket **160** may be placed on the cap plate **180** around the terminal hole **182**. The rivet **150** may be inserted into the terminal hole **182**.

[0073] The terminal coupling structure may be electrically connected to the electrode lead **92** or **95** through a lead terminal **170**. A lower insulation part **175** is disposed on a lower surface of the cap plate **180** to insulate the lead terminal **170** and the cap plate **180** from each other. The lower insulation part **175** may include a resin material substantially the same as or similar to the upper insulation part **140**.

[0074] Referring to FIG. 4, the rivet **150** may include a rivet body **150a** and a rivet wing **150b**. The gasket **160** may include a vertical extension part **160a** and a gasket wing **160b**.

[0075] The rivet body **150a** may refer to a portion which extends in a vertical direction (e.g., the third direction), and penetrates the terminal plate **130** and the cap plate **180**. The rivet wing **150b** may refer to a portion which extends in a horizontal direction (e.g., in the first direction) from an upper portion of the rivet body **150a**. The rivet wing **150b** may be formed as an integral member with the rivet body **150a**.

[0076] The vertical extension part **160a** of the gasket **160** extends in the third direction, and may be in contact with an outer circumferential surface of the rivet body **150a**. The vertical extension part **160a** may also be in contact with inner circumferential surfaces of the cap plate **180** and the lower insulation part **175**. The gasket wing **160b** may refer to a portion which is bent or extends in the horizontal direction (for example, in the first direction) from an upper portion of the vertical extension part **160a**. The gasket wing **160b** may be formed as an integral member with the vertical extension part **160a**.

[0077] The gasket wing **160b** may be sandwiched between the rivet wing **150b** and the cap plate **180**. According to exemplary embodiments, the rivet wing **150b** may be seated on the gasket wing **160b**. The rivet wing **150b** may be in contact with the upper surface of the gasket wing **160b** to be fastened to the cap plate **180** and the terminal plate **130**. Accordingly, the rivet **150** may be stably fixed to the electrode terminal parts **112** and **115**.

[0078] In addition, the vertical extension part **160a** may be in contact with the cap plate **180** and the lower insulation part **175** to surround the outer circumferential surface of the rivet body **150a**. Therefore, riveting stability may be further enhanced.

[0079] According to embodiments of the present disclosure, the rivet wing **150b** may include a first protrusion part **155**. The first protrusion part **155** may extend from a lower surface of the rivet wing **150b** to be inserted into an upper portion of the gasket wing **160b**.

[0080] The first protrusion part **155** may be integrally formed with the rivet wing **150b**, and may extend downward from the upper surface of the gasket wing **160b** in the third direction. The gasket wing **160b** may include a first recess into which the first protrusion part **155** is inserted.

[0081] According to the above-described embodiments, fusion stability or coupling stability of the

rivet **150** and the gasket **160** may be further reinforced by the first protrusion part **155**. Accordingly, inflow of external foreign substances occurring in the electrode terminal parts **112** and **115** and leakage of the electrolyte housed inside the case **110** may be prevented.

[0082] The first protrusion part **155** may press the gasket wing **160b** downward from the top of the gasket wing **160b**. Accordingly, the leakage of the electrolyte occurring at an interface of the gasket **160** and the cap plate **180** may be effectively prevented.

[0083] The upper insulation part **140** may be formed on the cap plate **180** to partially cover the gasket wing **160b**. The upper insulation part **140** may cover a portion of the gasket wing **160b** which protrudes outward from an outer circumferential surface of the rivet wing **150b** in the horizontal direction.

[0084] FIGS. **5** and **6** are partial enlarged cross-sectional views illustrating a terminal coupling structure of the cap assembly according to exemplary embodiments. FIG. **6** is an enlarged cross-sectional view illustrating the coupling of the rivet, the gasket and the cap plate of FIG. **5** (an area B of FIG. **3**).

[0085] Referring to FIG. **5**, the cap plate **180** may include a second protrusion part **185**. The second protrusion part **185** may extend from the upper surface of the cap plate **180** to be inserted into a lower portion of the gasket wing **160b**.

[0086] The second protrusion part **185** may be integrally formed with the cap plate **180**, and may extend upward from the lower surface of the gasket wing **160b** in the third direction. The gasket wing **160b** may include a second recess into which the second protrusion part **185** is inserted.

[0087] According to the above-described embodiments, the fusion stability or coupling stability of the gasket **160** and the cap plate **180** may be further enhanced by the second protrusion part **185**. Accordingly, the inflow of external foreign substances occurring in the electrode terminal parts **112** and **115** and the leakage of the electrolyte housed inside the case **110** may be more effectively prevented.

[0088] In some embodiments, the first and second protrusion parts **155** and **185** may press the gasket wing **160b** in both directions. In addition, the first and second protrusion parts **155** and **185** may function as teeth or screws that engage the gasket wing **160b**. Accordingly, the leakage of the electrolyte occurring at the interface of the gasket **160** and the cap plate **180** may be effectively prevented while improving the riveting stability.

[0089] Referring to FIG. **6**, a separation distance **W2** between the first protrusion part **155** and the rivet body **150a** may be about 30% or more of a length **W1** of the rivet wing **150b**. The separation distance **W2** of the first protrusion part **155** may be the shortest distance in the first direction from the outer circumferential surface of the rivet body **150a** to an inner circumferential surface of the first protrusion part **155**.

[0090] In some embodiments, the separation distance **W2** of the first protrusion part **155** may be about 40% or more, about 50% or more, or about 60% or more of the length **W1** of the rivet wing **150b**. In one embodiment, the separation distance **W2** of the first protrusion part **155** may be about 40% to 90%, about 50% to 90%, about 55% to 85%, or about 60% to 80% of the length **W1** of the rivet wing **150b**.

[0091] Within the above separation distance range, the blocking of the electrolyte leakage and the compression of the gasket wing **160b** through the first protrusion part **155** may be effectively implemented.

[0092] In some embodiments, a separation distance **L2** of the second protrusion part **185** from an outer circumferential surface of the gasket wing **160b** may be about 15% or more of a length **L1** of the gasket wing **160b**. For example, the separation distance **L2** of the second protrusion part **185** may be the shortest distance in the first direction from the outermost circumferential surface of the gasket wing **160b** to an outer circumferential surface of the second protrusion part **185**.

[0093] In some embodiments, the separation distance **L2** of the second protrusion part **185** may be about 20% or more, about 30% or more, about 40% or more, or about 50% or more of the length

L1 of the gasket wing **160b** (a length in the first direction). In one embodiment, the separation distance L2 of the second protrusion part **185** may be 20% to 90%, 30% to 90%, 40% to 80%, or 50% to 80% of the length L1 of the gasket wing **160b**.

[0094] Within the above separation distance range, the blocking of the electrolyte leakage and the compression of the gasket wing **160b** through the second protrusion part **185** may be effectively implemented.

[0095] In some embodiments, a height of the first protrusion part **155** (a height in the third direction) may be 0.1 mm to 1 mm. In one embodiment, the height of the first protrusion part **155** may be 0.1 mm to 0.5 mm. In one embodiment, the height of the first protrusion part **155** may be 0.1 mm to 0.4 mm, or 0.1 mm to 0.3 mm.

[0096] In some embodiments, a height of the second protrusion part **185** (a height in the third direction) may be 0.1 mm to 1 mm. In one embodiment, the height of the second protrusion part **185** may be 0.1 mm to 0.5 mm. In one embodiment, the height of the second protrusion part **185** may be 0.1 mm to 0.4 mm, or 0.1 mm to 0.3 mm.

[0097] Within the above-described height range of the protrusion part, a sufficient compressive force may be transmitted to the gasket wing **160b** while preventing mechanical damage to the gasket wing **160b**.

[0098] FIGS. **7** and **8** are partial enlarged cross-sectional views illustrating terminal coupling structures of the cap assembly according to exemplary embodiments.

[0099] Referring to FIG. **7**, the rivet wing **150b** may include a plurality of first protrusion parts **155**. In some embodiments, the cap plate **180** may also include a plurality of second protrusion parts **185**.

[0100] The plurality of first protrusion parts **155** are arranged in the first direction, and may be in contact with the upper portion of the gasket wing **160b**. The plurality of second protrusion parts **185** are arranged in the first direction, and may be in contact with the lower portion of the gasket wing **160b**.

[0101] In some embodiments, the first protrusion parts **155** and the second protrusion parts **185** may be alternately and repeatedly arranged in the first direction. Accordingly, screw coupling of the gasket wing **160b** may be substantially implemented by the protrusion parts **155** and **185**.

[0102] Referring to FIG. **8**, the rivet wing **150b** may include a plurality of first protrusion parts **155**. In some embodiments, the cap plate **180** may also include a plurality of second protrusion parts **185**.

[0103] Any one second protrusion part **185** of the second protrusion parts **185** may be overlapped with two or more first protrusion parts **155** in the height direction (third direction). In one embodiment, at least one other second protrusion part **185** of the second protrusion parts **185** may not be overlapped with the first protrusion part **155** in the height direction.

[0104] As described above, by imparting irregularity to the arrangement of the protrusion parts **155** and **185**, the possibility of electrolyte leakage may be further reduced.

[0105] FIGS. **9** to **11** are partial enlarged cross-sectional views illustrating terminal coupling structures of the cap assembly according to exemplary embodiments. FIGS. **9** to **11** are enlarged cross-sectional views illustrating the area B of FIG. **3**.

[0106] Referring to FIG. **9**, the protrusion parts **155** and **185** may have a trapezoidal shape, respectively. In some embodiments, the first protrusion part **155** and the second protrusion part **185** may have trapezoidal shapes which are mutually opposite phases.

[0107] Referring to FIG. **10**, the protrusion parts **155** and **185** may have a triangular shape, respectively. In some embodiments, the first protrusion part **155** and the second protrusion part **185** may have triangular shapes which are mutually opposite phases.

[0108] Referring to FIG. **11**, the protrusion parts **155** and **185** may have a curved shape, for example, a semicircular shape, respectively. In some embodiments, the first protrusion part **155** and the second protrusion part **185** may have semicircular shapes which are mutually opposite phases.

[0109] The shapes of the protrusion parts **155** and **185** may be appropriately changed in consideration of prevention of the electrolyte leakage and terminal coupling stability.

Claims

1. A secondary battery comprising: an electrode assembly which comprises a cathode and an anode; a case in which the electrode assembly is housed; and a cap assembly which covers the case, wherein the cap assembly comprises: a cap plate which covers the case and includes a terminal hole formed therein; a gasket disposed on the cap plate around the terminal hole; and a rivet which is inserted into the terminal hole to be coupled to the cap plate through the gasket, and includes a first protrusion part inserted into an upper portion of the gasket.
2. The secondary battery according to claim 1, wherein the rivet comprises a rivet body inserted into the terminal hole and a rivet wing extending from the rivet body in a horizontal direction, wherein the first protrusion part extends downward from a lower surface of the rivet wing.
3. The secondary battery according to claim 2, wherein a separation distance between the first protrusion part and the rivet body is 30% or more of a length of the rivet wing.
4. The secondary battery according to claim 2, wherein the gasket comprises a vertical extension part which is in contact with an outer circumferential surface of the rivet body, and a gasket wing extending from the vertical extension part in the horizontal direction, and wherein the first protrusion part is inserted into an upper portion of the gasket wing.
5. The secondary battery according to claim 4, wherein the cap plate comprises a second protrusion part inserted into a lower portion of the gasket wing.
6. The secondary battery according to claim 5, wherein the second protrusion part extends upward from an upper surface of the cap plate.
7. The secondary battery according to claim 5, wherein a plurality of the first protrusion parts and a plurality of the second protrusion parts are inserted into the gasket wing.
8. The secondary battery according to claim 7, wherein the first protrusion parts and the second protrusion parts are arranged alternately in the horizontal direction.
9. The secondary battery according to claim 7, wherein any one second protrusion part of the second protrusion parts is overlapped with two or more first protrusion parts in a vertical direction.
10. The secondary battery according to claim 5, wherein a distance of the second protrusion part from an outer circumferential surface of the gasket wing is 15% or more of a length of the gasket wing.
11. The secondary battery according to claim 4, wherein the cap assembly further comprises an upper insulation part disposed on the cap plate to partially cover the gasket wing.
12. The secondary battery according to claim 4, wherein the cap assembly further comprises a lower insulation part disposed on a lower surface of the cap plate to be in contact with the vertical extension part of the gasket.
13. A secondary battery comprising: an electrode assembly which comprises a cathode and an anode; a case in which the electrode assembly is housed; and a cap assembly which covers the case, wherein the cap assembly comprises: a cap plate which covers the case and includes a terminal hole formed therein; a gasket which is disposed on the cap plate around the terminal hole, and comprises a vertical extension part and a gasket wing bent from the vertical extension part in a horizontal direction; and a rivet which is coupled to the cap plate through the gasket, and comprises a rivet body inserted into the terminal hole and a rivet wing extending from the rivet body in the horizontal direction, wherein the rivet wing or the cap plate includes a protrusion part which is engaged to the gasket wing.
14. The secondary battery according to claim 13, wherein the gasket wing is sandwiched between the rivet wing and the cap plate.
15. The secondary battery according to claim 13, wherein the vertical extension part and the gasket

wing are a member integrally formed with each other.

16. The secondary battery according to claim 13, wherein the protrusion part includes a first protrusion part protruding downward from a lower surface of the rivet wing.

17. The secondary battery according to claim 13, wherein the protrusion part includes a second protrusion part protruding upward from an upper surface of the cap plate.
